CHB Inverters: Quiz and Exam Questions

EE561 Advanced Static Power Conversion · Multilevel Inverters · Nov 12, 2026 · @Elif

Part A: Post-presentation quiz

Closed book, 10 minutes, four multiple-choice questions with one correct answer each. No cheat sheet is needed. Q1 is numerical, Q2 conceptual, Q3 a design critique and Q4 checks the modulation trade-offs.

Q1. Unipolar PWM switching frequency

A single-phase H-bridge uses unipolar PWM with a triangular carrier of f_{cr} = 900 Hz and a 60 Hz sine reference. What are the device switching frequency $f_{sw,dev}$ and the inverter switching frequency $f_{sw,inv}$ (the pulse rate seen in v_{AB})?

- A. $f_{sw,dev}$ = 900 Hz, $f_{sw,inv}$ = 900 Hz
- B. $f_{sw,dev}$ = 900 Hz, $f_{sw,inv}$ = 1800 Hz
- C. $f_{sw,dev}$ = 1800 Hz, $f_{sw,inv}$ = 1800 Hz
- D. $f_{sw,dev}$ = 450 Hz, $f_{sw,inv}$ = 900 Hz

Q2. Unequal DC voltages

A CHB with two cells per phase at E and 2E gives seven levels, the same as three equal cells. Why does industry still rarely use unequal DC voltages?

- A. Unequal DC voltages give fewer output levels than equal ones for the same number of cells.
- B. Number of redundant switching states drops.
- C. Unequal DC voltages cannot be used with a three-phase load.
- D. The output voltage steps become larger than those of a two-level inverter.

Q3. Critique: when not to use a CHB

An engineer proposes a standard CHB drive, with each cell fed by a diode rectifier from the phase-shifting transformer, for a mine hoist. The hoist lowers heavy loads on every trip and must send the braking energy back to the grid. What is the main flaw?

- A. A CHB cannot run a motor at low speed, so the hoist cannot start.
- B. Diode rectifiers conduct only from the grid to the cells, so the braking energy cannot return to the grid.
- C. Phase-shifted PWM stops working when the power flow reverses.

- D. There is no flaw, the phase-shifting transformer returns the energy to the grid automatically.

Q4. Staircase modulation

Which of the following is an advantage of staircase modulation in a CHB?

- A. Each device switches only at the fundamental frequency, so switching losses are very low.
- B. Low-order harmonics stay small at every modulation index.
- C. The switching angles can change at any instant, giving a fast dynamic response.
- D. All cells always deliver equal power without any extra measures.

Quiz answer key (instructor only)

Q	Answer	Why
1	B	Each leg follows the 900 Hz carrier, so f_sw,dev = 900 Hz. The two legs switch at different instants (references 180° apart), so v_AB pulses twice per carrier period: f_sw,inv = 2 × 900 = 1800 Hz. Bipolar PWM would give A.
2	B	Different cells mean different designs and spare parts, and redundancy falls from 64 to 16 states, leaving little freedom for switching patterns. A is false: unequal cells give more levels per cell.
3	B	A diode-fed CHB has no path for power back to the grid. Regeneration needs an active front end in every cell, which is costly; CHB drives suit pumps, fans and compressors that rarely brake.
4	A	Fundamental-frequency switching gives the lowest switching losses, and the angles can be stored in a lookup table. B, C and D are its drawbacks: poor at low m_a, angles update once per cycle, unequal cell power.

Part B: Final exam questions

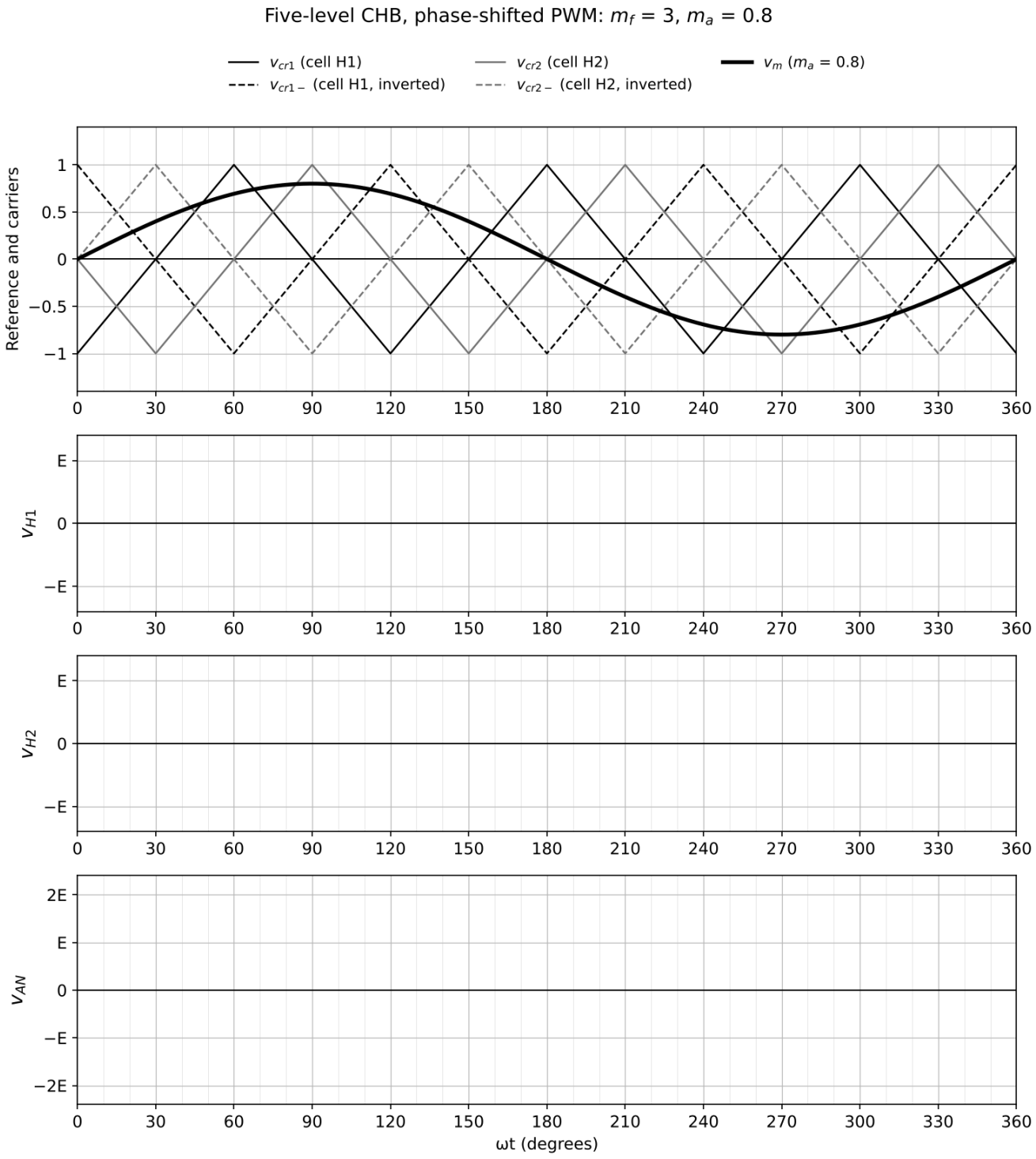
Three questions, 20 points each. Model solutions are in Part C.

Exam question 1. Phase-shifted PWM in a five-level CHB (20 pts)

A five-level CHB has H = 2 cells per phase, each with E = 1000 V. It uses phase-shifted unipolar PWM with a 50 Hz reference, f_{cr} = 150 Hz (m_f = 3) and m_a = 0.8. In cell k, S1k is on when $v_m > v_{crk}$ and S3k is on when $v_m < v_{crk-}$ (the inverted carrier); S4k and S2k are their complements. The cell output is v_{Hk} = +E when only S1k is on, −E when only S3k is on, and 0 otherwise.

1. Write the formula for the carrier phase shift φ_{cr} between the cells of a phase-shifted CHB, and evaluate it for this inverter. Check on the sheet that v_{cr2} is shifted from v_{cr1} by this angle. (3 pts)
2. On the sheet below, sketch v_{H1}, v_{H2} and $v_{AN} = v_{H1} + v_{H2}$ over one fundamental cycle. (9 pts)
3. Find the device switching frequency $f_{sw,dev}$ and the inverter switching frequency $f_{sw,inv}$. Using your sketch, explain why v_{AN} switches more often than either cell. (4 pts)
4. Find the peak fundamental phase voltage V_{AN1} and the rms fundamental line voltage V_{AB1} at m_a = 0.8. What is the largest V_{AB1} in the linear range? (4 pts)

Sketch sheet for Exam question 1



Exam question 2. True or false: PS-PWM, LS-PWM and staircase (20 pts)

Write T or F for each statement, then explain your answer in the box. An explanation is required for every statement, whether you mark T or F: show why it is true, or say what is wrong and give the correct statement. An answer without a correct explanation gets no credit. All statements carry equal weight.

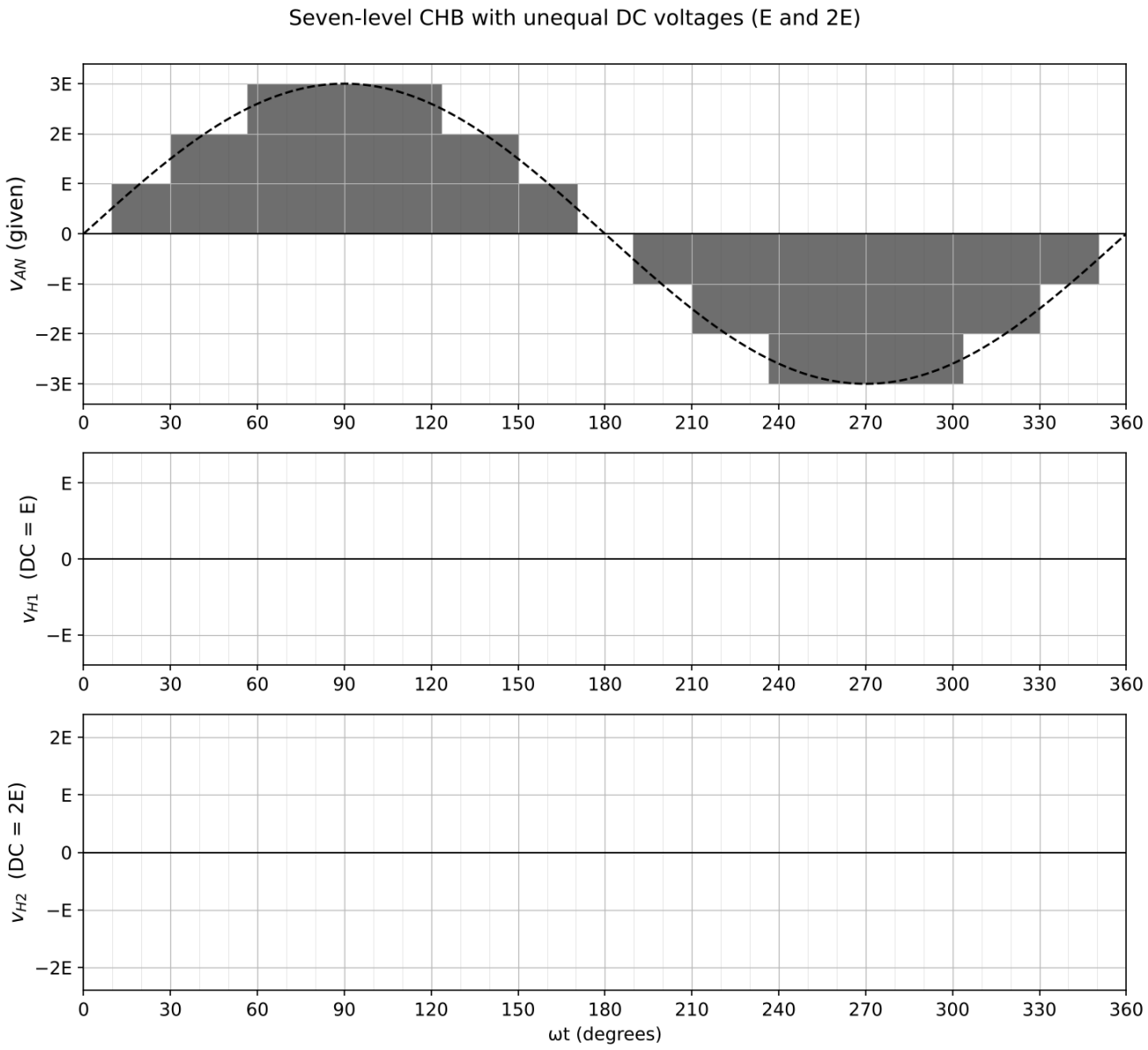
#	Statement	T / F	Explanation (required)
1	A single H-bridge cell can output three voltage levels: +E, 0 and −E.		
2	A CHB with four equal cells per phase gives a nine-level phase voltage.		
3	With bipolar PWM, the two legs of an H-bridge switch independently, so v_AB has three levels.		
4	A three-phase five-level CHB uses 12 switches in total.		
5	A CHB whose cells are fed by diode rectifiers can send braking energy back to the grid without extra hardware.		
6	Staircase modulation has very low switching losses because each device switches only at the fundamental frequency.		
7	Staircase modulation keeps a good waveform at low modulation index because all cells keep switching.		

Exam question 3. CHB with unequal DC voltages (20 pts)

One phase of a seven-level CHB has cell H1 with DC voltage E and cell H2 with 2E. The sheet below gives the staircase v_AN, which steps at θ = 9.6°, 30° and 56.4° in each quarter cycle. Use these states: ±E is made by H1 alone (v_H2 = 0), ±2E by H2 alone (v_H1 = 0), and ±3E by both cells.

1. On the sheet, sketch v_H1 and v_H2 over one fundamental cycle. (8 pts)
2. Give two advantages? (4 pts)
3. Give two disadvantages of unequal DC voltages that explain their limited use in industry. (4 pts)

Sketch sheet for Exam question 3



Part C: Model solutions (instructor only)

Exam question 1

1. φ_cr = 360° / (m − 1) = 360° / 4 = 90° (equivalently 180° / H). The inverted carriers v_cr1− and v_cr2− are 180° from v_cr1 and v_cr2.
2. See the solution sheet below. Each cell gives a three-level (unipolar) pulse train; v_AN steps by one level E at a time and reaches ±2E only near the peaks.
3. f_sw,dev = f_cr = 150 Hz. f_sw,inv = 2H · f_sw,dev = (m − 1) · f_sw,dev = 4 × 150 = 600 Hz. The shifted carriers make the two cells switch at different instants, so every transition of both cells appears separately in v_AN.
4. V_AN1 = m_a · H · E = 0.8 × 2000 = 1600 V peak. V_AB1 = √3 × 1600 / √2 = 1960 V rms. At m_a = 1: 2449 V rms (2.83 kV with third-harmonic injection).

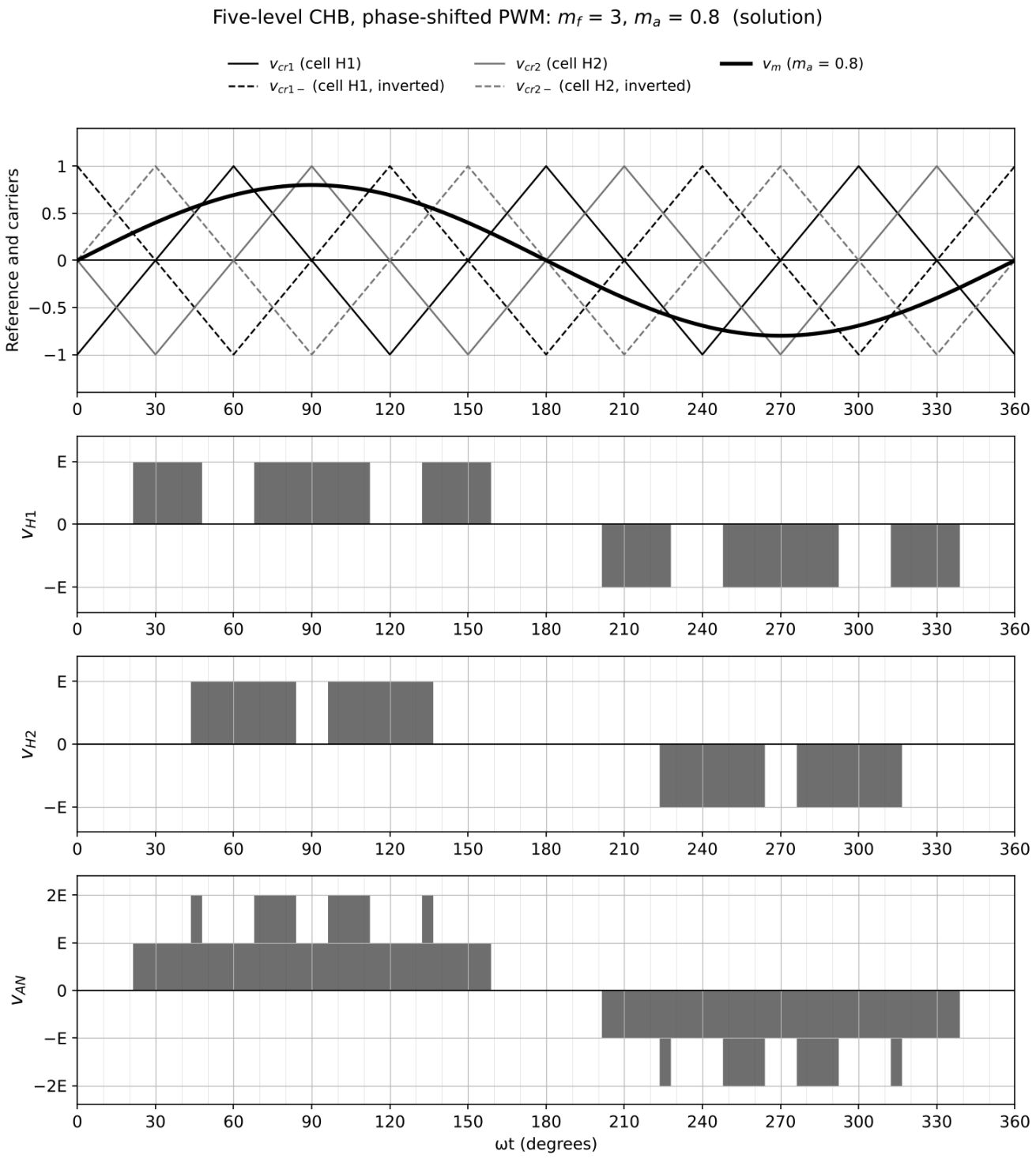
Exam question 2

#	Answer	Expected explanation
1	T	+E (S1, S2 on), 0 (two zero states) and −E (S3, S4 on).
2	T	m = 2H + 1 = 2(4) + 1 = 9.
3	F	That is unipolar PWM. With bipolar PWM the legs are complementary and v_AB has only two levels, +E and −E.
4	F	N_sw = 6(m − 1) = 6 × 4 = 24 switches.
5	F	Diode rectifiers conduct only from the grid to the cells; regeneration needs an active front end in every cell.
6	T	Each cell turns on and off once per half cycle, so switching losses are very low.
7	F	At low m_a the outer cells stop switching, only the inner levels are used, and low-order harmonics grow.

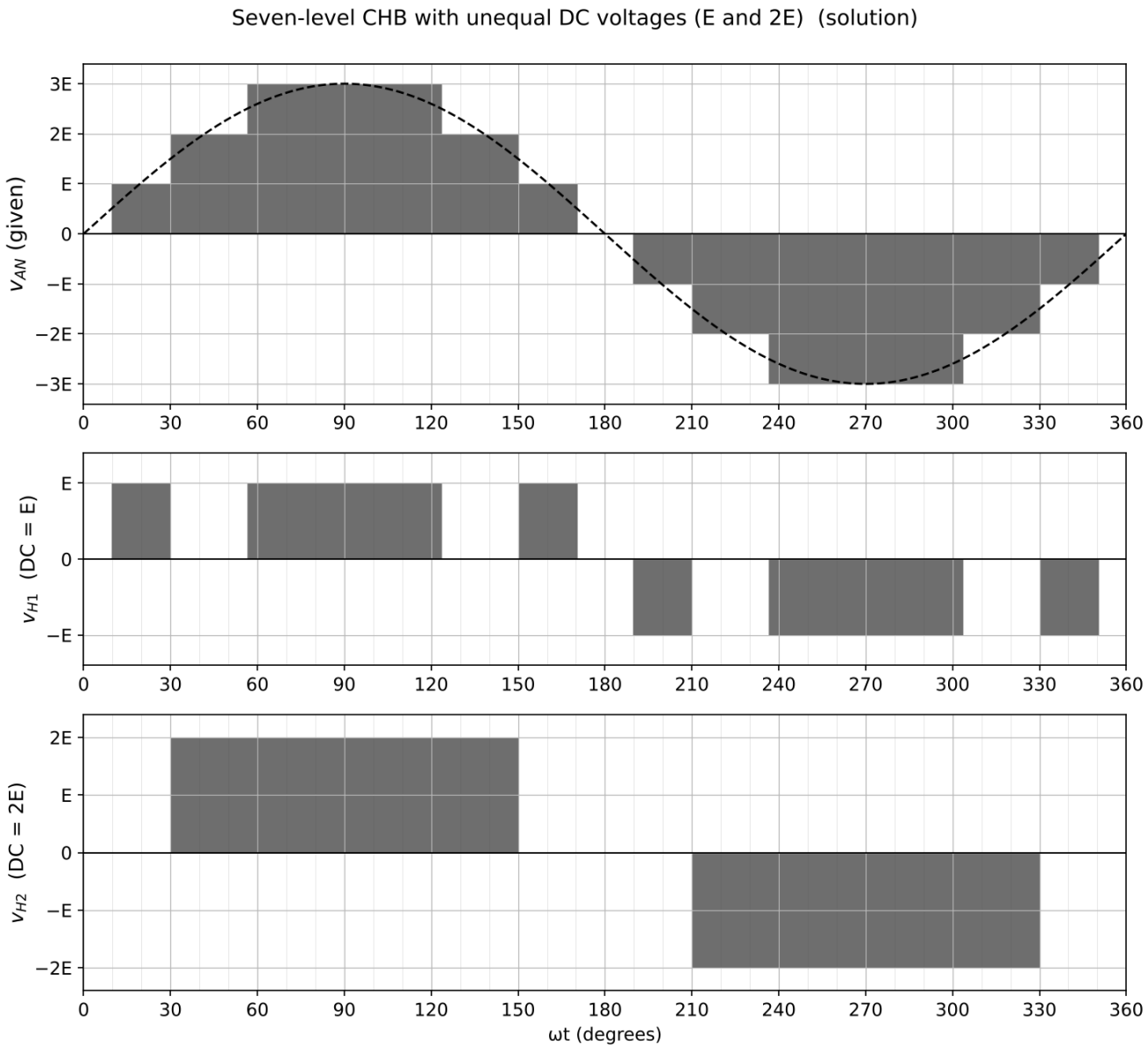
Exam question 3

1. See the solution sheet below. H1 is on for |v_AN| = E and 3E; H2 is on for |v_AN| ⩾ 2E (30° to 150° and 210° to 330°).
2. Any two: more levels from fewer cells (7 levels from two cells instead of three; 9 levels with E and 3E instead of four cells); fewer switches, gate drivers and transformer secondaries for the same number of levels; the larger cell switches only 4 times per cycle, so its switching loss is low.
3. Any two: cells are no longer identical, so modularity and spare-part sharing are lost; redundancy drops from 64 to 16 switching states, leaving little freedom in switching patterns; H2 needs devices rated for 2E and carries more power and losses, so the thermal design is uneven; the transformer needs secondaries at two voltages.

Solution sheet for Exam question 1



Solution sheet for Exam question 3


Seven-level CHB with unequal DC voltages (E and 2E) (solution)
v_{AN} (given)
3E
2E
E
0
−E
−2E
−3E
v_{H1} (DC = E)
v_{H2} (DC = 2E)
0
30
60
90
120
150
180
210
240
270
300
330
360
ωt (degrees)